CSA1606 - DATA WAREHOUSING AND DATA MINING

CO4_ ASSIGNMENT TOOL – 2

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CO4: Applicability of Clustering Algorithms in Real-World Scenarios

Segmentation, Fraud Detection, Churn Analysis, Recommendation Systems & Traffic Management

Q1. Customer Segmentation — Supermarket Purchases

Using the dataset below, propose a clustering approach to segment customers and explain how it supports targeted marketing. (BL3 – 7 Marks)

Customer	Visits/Month	Avg Spend (■)	Items Bought
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Proposed Approach: K-Means Clustering

K-Means is suitable here because the dataset has small, continuous, numeric features (visits, spend, items bought) and no labelled classes exist beforehand. The steps are:

- **Data preparation:** Standardize the three features (Z-score normalization) since "Avg Spend" is on a much larger numeric scale (■1800–■6000) than "Visits" or "Items Bought"; without scaling, spend would dominate the distance calculation.
- **Choosing k:** Using the Elbow Method / Silhouette Score on this small dataset suggests k = 2 clusters, since the data visibly splits into two natural groups.
- **Cluster assignment (based on the data):** Cluster A = {C1, C3, C5} — frequent visitors, low average spend per visit, high items bought ("Frequent Bulk Buyers"). Cluster B = {C2, C4} — infrequent visitors, high average spend, fewer items bought ("High-Value Occasional Shoppers").

- **Validation:** Compute within-cluster sum of squares (WCSS) and inspect cluster centroids to confirm the two groups are well separated and internally consistent.

How this Supports Targeted Marketing

- **Frequent Bulk Buyers (C1, C3, C5):** Target with loyalty programs, monthly combo offers, and bulk-purchase discounts to increase basket size further.
- **High-Value Occasional Shoppers (C2, C4):** Target with premium product recommendations, personalized re-engagement campaigns (SMS/email) to increase visit frequency, and cross-sell offers.
- Marketing budget is allocated cluster-wise instead of uniformly, improving ROI and customer retention while reducing irrelevant promotions.

Q2. Fraud Detection — Credit Card Transactions

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. (BL4 – 7 Marks)

Transaction	Amount (■)	Location	Time (hrs)
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Proposed Approach: DBSCAN (Density-Based Clustering)

Unlike K-Means, fraud detection needs an algorithm that does not force every point into a cluster and can flag outliers explicitly — DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is ideal because fraud is rare and does not follow a fixed cluster shape.

Design Steps

- **Feature encoding:** One-hot encode "Location" and scale "Amount" and "Time" so all features contribute proportionally to distance.

- **Apply DBSCAN:** Set parameters ε (neighbourhood radius) and MinPts (minimum points to form a dense region) based on typical transaction behaviour.
- **Dense cluster (normal behaviour):** T1, T3, T5 — small amounts (■150–■200), same location (Chennai), daytime hours (10–12 hrs) — form one dense, tightly-packed cluster.
- **Noise points (potential fraud):** T2 (■5000, Delhi, 2 hrs) and T4 (■7000, Mumbai, 1 hr) lie far from the dense cluster in both amount and odd late-night/early-morning timing — DBSCAN labels these as noise/outliers rather than forcing them into a cluster.
- **Flagging:** All points labelled "noise" (cluster label = -1) are automatically flagged for manual review or automated blocking.

Why This Works for Fraud Detection

- Fraudulent transactions are rare and irregular — they do not need a predefined cluster count (unlike K-Means, DBSCAN does not require specifying k).
- High amount + unusual location + odd transaction hour together create a low-density region, which DBSCAN naturally isolates as an anomaly.
- The bank can trigger real-time alerts, temporary holds, or OTP verification specifically for transactions falling in low-density/noise regions.

Q3. Churn Prediction — Telecom Usage

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. (BL4 – 7 Marks)

User	Call Duration (min)	Data Usage (GB)	Recharge (■)
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Proposed Approach: K-Means Clustering

K-Means is applied on standardized usage features (call duration, data usage, recharge amount) to segment users into behaviourally similar groups.

Resulting Clusters (from the data)

- **Cluster 1 — High Engagement Users {U1, U3, U5}:** High call duration (300–350 min), high data usage (5–6 GB), high recharge value (■200–220) → low churn risk, high revenue contribution.
- **Cluster 2 — Low Engagement Users {U2, U4}:** Low call duration (80–100 min), low data usage (1–2 GB), low recharge (■90–100) → high churn risk, minimal engagement with services.

Application to Churn-Prone Identification & Retention

- **Early warning signal:** Users whose usage metrics drift from the "High Engagement" cluster toward the "Low Engagement" cluster over successive months are flagged as churn-prone before they actually leave.
- **Targeted retention offers:** Low-engagement cluster users can be proactively offered discounted recharge plans, bonus data, or loyalty points to re-engage them.
- **Resource prioritization:** Customer-care teams and retention budgets focus on the cluster at risk, rather than treating the entire customer base uniformly, increasing retention efficiency and reducing churn-related revenue loss.
- **Personalized plans:** High-engagement clusters can be up-sold premium plans, while low-engagement clusters get win-back campaigns tailored to their specific usage gap.

Q4. Recommendation Systems — E-commerce Behaviour

Suggest a clustering technique to group users and explain how it improves recommendation systems. (BL3 – 7 Marks)

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Proposed Approach: K-Means Clustering

Since the three behavioural features (views, clicks, purchases) are numeric, continuous, and strongly correlated with purchase intent, K-Means clustering can group users by engagement level after normalizing the features.

Resulting Clusters (from the data)

- **Cluster 1 — Highly Active Buyers {U1, U3, U5}:** High product views (50–60), high clicks (20–25), high purchases (5–6) → strong purchase intent.
- **Cluster 2 — Passive Browsers {U2, U4}:** Low views (20–30), low clicks (8–10), low purchases (1–2) → weak purchase intent, mostly window-shopping.

How this Improves Recommendation Systems

- **Cluster-based collaborative filtering:** Recommendations for a user are drawn from the purchase/view patterns of other users in the same cluster, improving relevance compared to a one-size-fits-all recommendation list.
- **Highly Active Buyers:** Shown complementary/upsell products, limited-time deals, and "frequently bought together" items to convert further engagement into sales.
- **Passive Browsers:** Shown broader, top-trending or discounted products, and re-targeting banners designed to increase clicks and move them toward the buyer cluster.

- **Reduced cold-start impact:** New users can be assigned to a cluster based on early browsing signals, immediately benefiting from that cluster's recommendation pattern.
- Overall, clustering personalizes recommendations at the segment level, improving click-through rate (CTR) and conversion rate while reducing irrelevant suggestions.

Q5. Traffic Congestion Zones — Traffic Data

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. (BL4 – 7 Marks)

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Proposed Approach: K-Means Clustering

Vehicle count, average speed, and delay are numeric features with a clear inverse relationship (high vehicle count → low speed → high delay), making K-Means effective for separating congested from free-flowing locations after feature

Resulting Clusters (from the data)

- **Cluster 1 — Congested Zones {L2, L4}:** High vehicle count (500–600), low average speed (10–15 km/h), high delay (25–30 min) → severe congestion.
- **Cluster 2 — Free-Flowing Zones {L1, L3, L5}:** Moderate vehicle count (200–220), high average speed (28–32 km/h), low delay (9–12 min) → smooth traffic movement.

Application to Traffic Management

- **Real-time signal control:** Congested-zone locations (L2, L4) receive dynamic signal timing adjustments and longer green-light durations to ease vehicle build-up.

- **Resource deployment:** Traffic police, tow trucks, and monitoring cameras are prioritized at congested-cluster locations rather than spread evenly across the city.
- **Route diversion & navigation apps:** Congested zones are flagged to commuters via navigation apps (e.g., Google Maps-style alerts), suggesting alternate routes through free-flowing zones.
- **Infrastructure planning:** Repeated congestion in the same cluster over time justifies long-term interventions such as flyovers, lane widening, or new public transport routes.
- Clustering therefore converts raw sensor data into actionable, location-specific traffic management decisions instead of a uniform city-wide policy.